

So volume of liquid increases by $0.920 - 0.893 = 0.03 \text{ cm}^3$

7.

Ans: D

work done required by the motor = net potential gain by the system
= $m_1gx - m_2gx$, where x is the distance raised and fallen by m_1 and m_2 respectively.

Hence the rate the motor provides energy is $(m_1 - m_2)gv$

8.

Ans: B